Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

Student Name (print): _____

This exam contains 5 pages (including this cover page) and 5 questions. The total number of possible points is 15. Enter your answers in the space provided. Write your final answer on the

"Solution" line for each problem, where appropriate. Otherwise, draw a box around your final answer. Complete your solutions to the "show your work" problems on the page indicated.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit**. A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations may still receive partial credit.
- **Provide exact answers** unless otherwise instructed.
- **Simplify all answers as much as possible.** This means that you need to need to combine like terms, reduce fractions, etc. (You do not need to rationalize denominators.)
- **Be sure to state units for applied problems.**
- **Clearly identify your answer for each problem.**

Do not write in the table to the right.

Question	Points	Score
1	4	
2	2	
3	2	
4	3	
5	4	
Total:	15	

Points earned: _____ out of 0 points

1. (4 points) Let $f(x)$ be the following expression, $f(x) = 48x + 14$. Determine if $f(x)$ is an element of the set of polynomials.

   YES NO

   If you circled "YES," write a paragraph indicating why $f(x)$ is an element of the set of polynomials. If you circled "NO," write a paragraph indicating why $f(x)$ is not an element of the set of polynomials.

2. (2 points) Find the zeros of $f(x) = x^3 - x$, then construct a rough graph of $f(x)$.

3. (2 points) Find the zeros of $g(x) = x^2 - 16$ and then construct a rough graph of $g(x)$.

Points earned: _____ out of 8 points

4. (3 points) What is $5 \cdot 3$?
   A. 8 B. 15 C. 2 D. none of the above

4. _____

5. (4 points) What is $\frac{\partial z}{\partial t}$ and $\frac{\partial z}{\partial s}$ for the equation $z = t \cdot s$.

   ◯ t

   ◯ s

   ◯ z

   ◯ I don't know

Points earned: _____ out of 7 points